



Power BI Ultimate Cheat Sheet



DAX Essentials & Advanced



Aggregation

```
SUM(Sales[Amount])  
AVERAGE(Orders[Quantity])  
COUNT(Customer[ID])  
DISTINCTCOUNT(User[ID])  
MAX(Revenue[Total])  
MIN(Inventory[Stock])
```



Time Intelligence

```
TOTALYTD(Sales[Amount], Dates[Date])  
DATEADD(Dates[Date], -1, MONTH)  
SAMEPERIODLASTYEAR(Dates[Date])  
PARALLELPERIOD(Dates[Date], -1, QUARTER)
```



Logical & Conditional

```
IF(Sales[Amount] > 1000, "High", "Low")  
SWITCH(Month, 1, "Jan", 2, "Feb", "Other")  
AND(Condition1, Condition2)  
OR(Condition1, Condition2)  
NOT(Condition)
```



Text & Formatting

```
CONCATENATE(FirstName, " ", LastName)  
FORMAT(Sales[Amount], "Currency")  
LEFT(Product[Code], 3)  
SEARCH("A", Name)  
SUBSTITUTE(Name, "Inc", "")
```



Advanced DAX

```
CALCULATE(SUM(Sales[Amount]), FILTER(Sales, Sales[Region] = "West"))  
VAR Total = SUM(Sales[Amount])  
RETURN Total * 0.1
```



Table Functions

```
SUMMARIZE(), ADDCOLUMNS(), SELECTCOLUMNS(), GENERATE(), CROSSJOIN()
```

Data Modeling Best Practices

-  Use star schema: Fact table + dimension tables
-  Avoid bi-directional relationships unless necessary
-  Implement Row-Level Security (RLS) with roles
-  Use measures over calculated columns for performance
-  Use hierarchies for drill-down (e.g., Year > Quarter > Month)
-  Use descriptive names for fields and tables
-  Hide unused columns from report view

Visualization Tips

Tip	Benefit
Use slicers and filters	Enhance interactivity
Use tooltips and drill-through	Provide context
Apply conditional formatting	Highlight trends
Use bookmarks and buttons	Create guided navigation
Group visuals and use themes	Improve aesthetics
Use KPI cards for key metrics	Quick insights
Use custom visuals from marketplace	Expand capabilities

Power BI Desktop Shortcuts

Shortcut	Action
Ctrl + N	New report
Ctrl + S	Save
Ctrl + Shift + C	Copy visual as image
Ctrl + D	Duplicate visual
Ctrl + Alt + B	Open bookmarks pane
Alt + Shift + F	Focus mode for visual
Ctrl + T	Add text box
Ctrl + K	Add hyperlink

Performance Optimization

- Use Performance Analyzer to debug slow visuals
- Avoid complex calculated columns in large tables
- Use aggregated tables for large datasets
- Limit visuals per page to reduce rendering time
- Use variables in DAX for cleaner logic

- Use summary tables for high-level dashboards

Power Query Tips

- Use Remove Columns to reduce model size
- Use Group By for aggregations
- Use Merge Queries for joins
- Use Custom Columns with M code:

M

```
if [Sales] > 1000 then "High" else "Low"
```

- Use Query Dependencies view to track data flow
- Enable query folding for performance

Integrations

Tool	Integration
Excel	Analyze in Excel, export to .xlsx
Python/R	Run scripts in Power Query or visuals
Azure Synapse	Connect to big data
SharePoint/OneDrive	Auto-refresh from cloud files
Power Automate	Trigger workflows from report actions
Teams	Share reports directly in chat or channels
SQL Server	DirectQuery or import mode

Security & Governance

Feature	Description
Row-Level Security (RLS)	Restrict data by user role
Workspace roles	Admin, Member, Contributor, Viewer
Data sensitivity labels	Tag reports with confidentiality levels
Audit logs	Track report access and changes
Deployment pipelines	Manage dev → test → prod environments